

TEJAS VENU

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EDUCATION

University of Virginia

Master of Science in Computer Science - GPA: 4.00 / 4.00

Aug 2025 – Dec 2026

Charlottesville, VA

Visvesvarya Technological University

Bachelor of Engineering in Computer Science - GPA: 3.90 / 4.00

Dec 2021 – Jul 2025

Bengaluru, India

WORK EXPERIENCE

Internship - AI/ML Developer

Oracle

Feb 2025 – Jul 2025

Bengaluru, India

- Automated extraction of key legal clauses from **100+ page** documents using **LLMSherpa**, **Word2Vec** embeddings and **dense vector search** across **8 US regions**.
- Improved retrieval accuracy from **61% to 93%** by designing a **LLM-based (Cohere Command R+) taxonomy filtering layer**, eliminating irrelevant semantic matches at scale.
- Led a **5-member team** in the design of an LLM-based taxonomy filtering layer, which played a key role in lowering annual client costs from **\$2.3M** to **\$200K–\$300K**.

Internship - Data Analyst

Toyota Kirloskar Auto Parts

Jun 2024 – Sep 2024

Bengaluru, India

- Automated Excel-based safety validations for large datasets (**1000 rows × 700 columns**) using **Python** and **Pandas**, replacing manual checks driven by frequently changing criteria sheets.
- Reduced execution time from **140s to 20s** by eliminating Excel Macros and implementing an optimized Python pipeline.
- Built a **Tkinter** GUI for one-click execution, enabling fully automated validation for gears, axles and driveshafts with **100% accuracy** and zero manual intervention.

PROJECTS

Explainable Skin Lesion Classification with Attention-Enhanced CNNs | Python, PyTorch, EfficientNet-B0

- Built an interpretable skin-lesion classification model using **EfficientNet-B0** with a **metadata-conditioned spatial attention** module, embedding explainability directly into training.
- Achieved **87.4%** test accuracy and improved macro-F1 from **0.776 → 0.789**, delivering balanced performance across **7 lesion classes** on the **HAM10000** dataset.
- Replaced post-hoc **Grad-CAM** with **trainable attention** and validated robustness through ablation studies, demonstrating improved interpretability without compromising performance.

Human Intruder Detection System | Python, PyTorch, Raspberry Pi

- Built a real-time intruder detection system using **YOLO**, **MTCNN** and **ResNet**, classifying students, faculty, and intruders from live CCTV feeds.
- Achieved **95%+ accuracy** by combining face recognition with uniform and ID-based classification on an augmented dataset.
- Enabled **low-latency, real-time alerts** through integration with existing CCTV infrastructure, ensuring reliable performance across varied lighting and camera angles.

SKILLS

Programming: Python, Java, C, C++

AI / ML & Data Science: PyTorch, TensorFlow, Keras, Pandas, NumPy, LangChain, Hugging Face

Web Development: HTML, CSS, JavaScript, NodeJS, React, Streamlit, Tkinter, Django

Databases: PostgreSQL, MySQL

Version Control & Analytics Tools: Git, GitHub, Power BI, Matplotlib, Seaborn